Veganism, bone mineral density, and body composition: a study in Buddhist nuns.

SUMMARY: This cross-sectional study showed that, although vegans had lower dietary calcium and protein intakes than omnivores, veganism did not have adverse effect on bone mineral density and did not alter body composition. INTRODUCTION: Whether a lifelong vegetarian diet has any negative effect on bone health is a contentious issue. We undertook this study to examine the association between lifelong vegetarian diet and bone mineral density and body composition in a group of postmenopausal women. METHODS: One hundred and five Mahayana Buddhist nuns and 105 omnivorous women (average age = 62, range = 50-85) were randomly sampled from monasteries in Ho Chi Minh City and invited to participate in the study. By religious rule, the nuns do not eat meat or seafood (i.e., vegans). Bone mineral density (BMD) at the lumbar spine (LS), femoral neck (FN), and whole body (WB) was measured by DXA (Hologic QDR 4500). Lean mass, fat mass, and percent fat mass were also obtained from the DXA whole body scan. Dietary calcium and protein intakes were estimated from a validated food frequency questionnaire. RESULTS: There was no significant difference between vegans and omnivores in LSBMD (0.74 +/- 0.14 vs. 0.77 +/- 0.14 g/cm(2); mean +/- SD; P = 0.18), FNBMD (0.62 +/- 0.11 vs. 0.63 +/- 0.11 g/cm(2); P = 0.35), WBBMD (0.88 +/- 0.11 vs. 0.90 +/- 0.12 g/cm(2); P = 0.31), lean mass (32 +/- 5 vs. 33 +/- 4 kg; P = 0.47), and fat mass (19 +/- 5 vs. 19 +/- 5 kg; P = 0.77) either before or after adjusting for age. The prevalence of osteoporosis (T scores < or = -2.5) at the femoral neck in vegans and omnivores was 17.1% and 14.3% (P = 0.57), respectively. The median intake of dietary calcium was lower in vegans compared to omnivores (330 +/- 205 vs. 682 +/- 417 mg/day, P < 0.001); however, there was no significant correlation between dietary calcium and BMD. Further analysis suggested that whole body BMD, but not lumbar spine or femoral neck BMD, was positively correlated with the ratio of animal protein to vegetable protein. CONCLUSION: These results suggest that, although vegans have much lower intakes of dietary calcium and protein than omnivores, veganism does not have adverse effect on bone mineral density and does not alter body composition.